

Greedy Coloring Algorithm

Greedy coloring or sequential coloring is a coloring of the vertices of a graph formed by a greedy algorithm that considers the vertices of the graph in sequence and assigns each vertex its first available color.

Step I: Choose an order for the vertices say $v_1, v_2, \dots, v_n$ and also

choose the list of colors in a particular order say $1, 2, 3, \dots, n$

Step II: Color the vertex v_i with the first available color and then in the i th step assign to v_i the smallest indexed color not already used for its neighbour

among, $v_1, v_2, \dots, v_{i-1}$

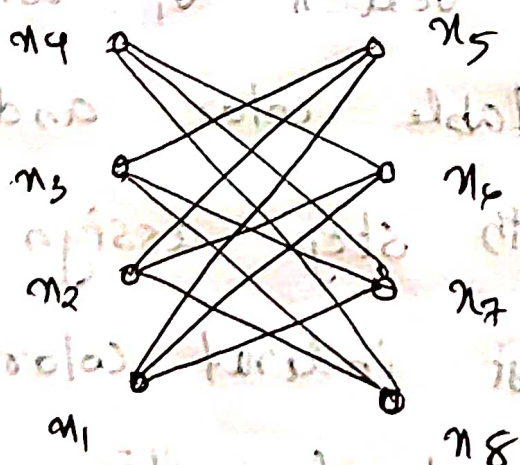
Continue until the last vertex v_n is

Note:

1. In a vertex coloring each vertex has at most $d_{\max}$ (maximum degree of the vertex) earlier neighbours. So Greedy algorithm cannot force to use more than $d_{\max} + 1$ colors.

2. Changing the ordering of vertices might change the number of colors required.

Eg: Color the following graph by Greedy coloring algorithm.



Choose an arbitrary ordering of

$v_1, v_2, v_3, v_4, v_5, v_6, v_7, v_8$

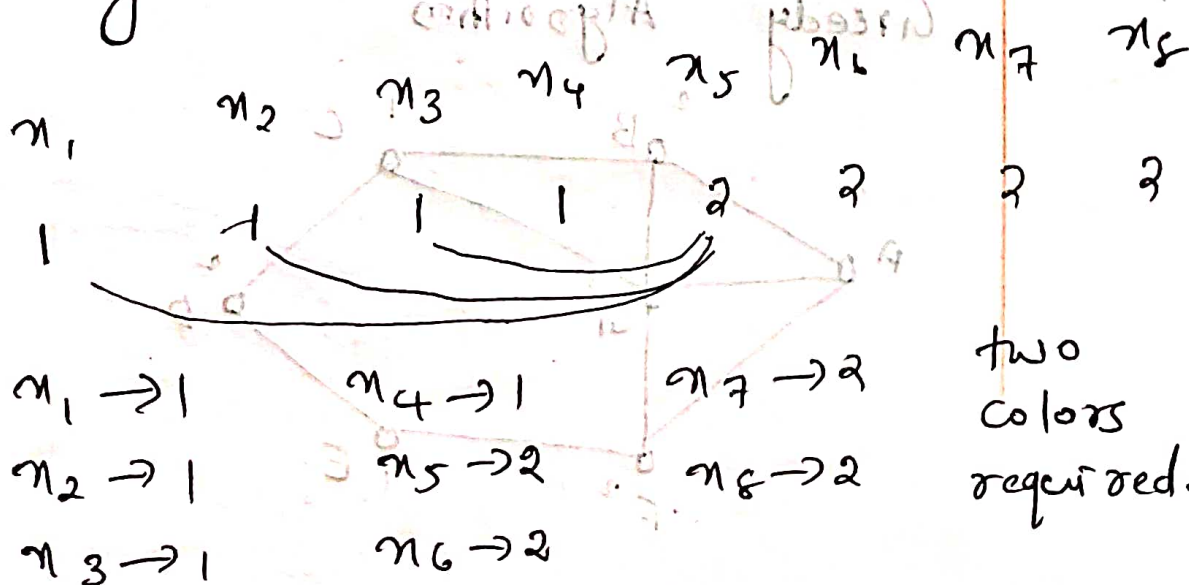
Let the colors be chosen in the order, 1, 2, 3,

color v_1 with color 1. Now to color v_2 , choose the first available color such that (or the smallest indexed color not used to color the vertices on L.H.S which are incident to v_2).

There are no neighbours for v_2 on L.H.S. So color v_2 with color 1.

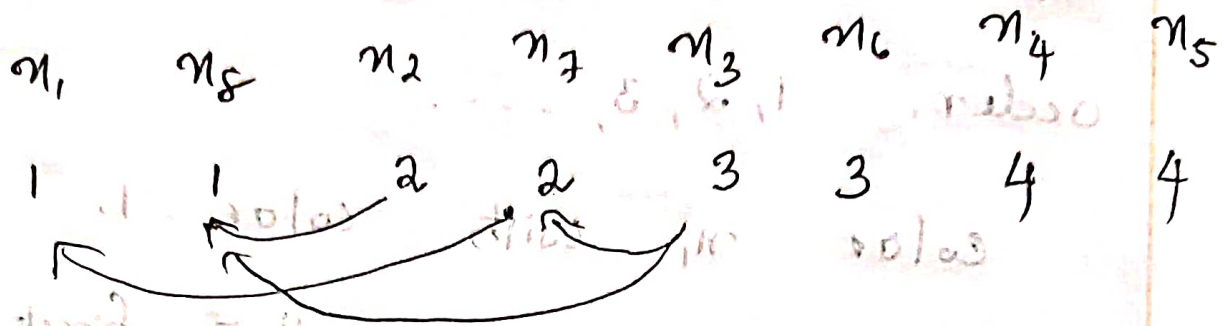
Now to color v_3 we use the similar

criteria, following this we get the coloring as.



Considering another ordering for

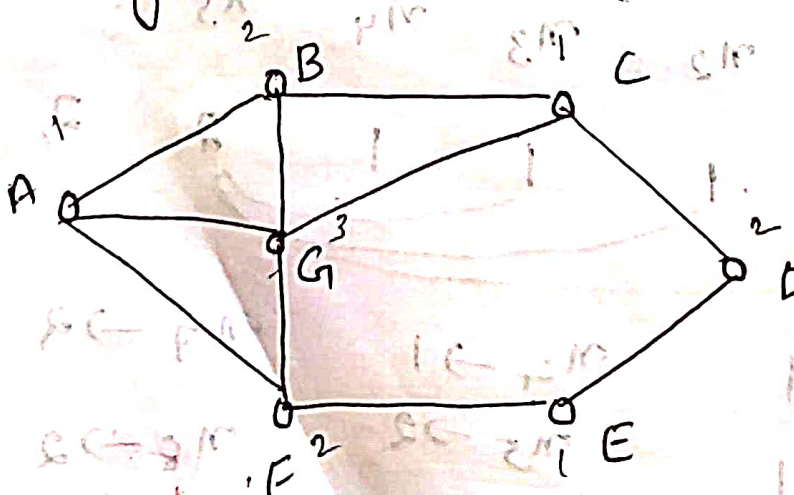
the above graph.



Here as the ordering changed the number of colors required became 4.

$n_1 \rightarrow 1$ $n_4 \rightarrow 4$ or $n_7 \rightarrow 2$
 $n_2 \rightarrow 2$ $n_5 \rightarrow 4$
 $n_3 \rightarrow 3$ $n_6 \rightarrow 3$ $n_8 \rightarrow 1$

(2) Color the following graph by Greedy Algorithm



Let the ordering be

A, B, C, D, E, F, G & colors be

1, 2, 3, ...

Now.

A	B	C	D	E	F	G
1	3	1	2	1	2	3

C has neighbour B on L.H-S ~~side~~

So we can use color 1.